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peppwR: Simulation-Based Power Analysis for Phosphoproteomics Experiments in R

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Abstract

Phosphoproteomics experiments require careful experimental design to ensure adequate statistical power, yet no dedicated tools exist for power analysis that account for the unique statistical properties of peptide-level quantification data. We present **peppwR**, an R package for simulation-based power analysis tailored to phosphoproteomics. **peppwR** fits distributions to pilot data on a per-peptide basis, capturing the heterogeneity in abundance and variance across the peptidome. It supports multiple statistical tests (Wilcoxon, bootstrap-t, Bayes factor), tracks missingness patterns common in mass spectrometry data, and provides FDR-aware power calculations. Users can determine required sample sizes, estimate power for planned experiments, or identify minimum detectable effect sizes.

Keywords: power analysis, phosphoproteomics, mass spectrometry, experimental design, R.

1. Introduction

Statistical power—the probability of correctly rejecting a false null hypothesis—represents one of the most fundamental concepts in experimental design (Cohen 1988). Introduced by Neyman and Pearson (1933) and formalized by Cohen (1962), power analysis provides the theoretical framework for determining appropriate sample sizes, estimating the likelihood of detecting true effects, and identifying minimum detectable effect sizes. The consequences of inadequate power are profound: under-powered studies waste resources and risk missing genuine biological discoveries, while over-powered designs are economically inefficient and may detect biologically trivial effects (Button, Ioannidis, Mokrysz, Nosek, Flint, Robinson, and Munafò 2013; Ioannidis 2005).

Despite widespread adoption in clinical trials (Schulz and Grimes 2005) and genomics (Hart, Therneau, Zhang, Poland, and Kocher 2013), power analysis remains systematically neglected in proteomics research. Recent surveys indicate that fewer than 15% of published proteomics studies report formal power calculations (Wilkinson 2019), compared to over 80% in clinical research (Bacchetti 2002). This disparity persists despite the field's evolution toward increasingly quantitative approaches and the availability of sophisticated statistical methods for proteomics data analysis (Karp, Huber, Sadowski, Charles, Hester, and Lilley 2010).

Mass spectrometry-based proteomics presents unique statistical challenges that violate fundamental assumptions of traditional power analysis. First, peptide abundances exhibit extreme heterogeneity, with concentrations spanning 6-8 orders of magnitude within a single experiment (Olsen, Blagoev, Gnad, Macek, Kumar, Mortensen, and Mann 2006; Humphrey, Azimifar, and Mann 2015). This heterogeneity extends beyond mean abundances to encompass dramatic differences in variance structure across the peptidome (Karp *et al.* 2010; Varemo, Henriksen, Scheele, Broholm, Pedersen, Uhlén, Pedersen, and Nielsen 2013). Second, abundance distributions deviate substantially from normality, typically exhibiting right skewness better characterized by gamma, lognormal, or inverse Gaussian distributions (Hultin, Hausner, Hultin, and Giorgi 2005; Webb-Robertson, Wiber, Matzke, Samuel, Rodland, Clauss, and Pounds 2015). These distributional properties reflect both biological variation and the multiplicative nature of mass spectrometry quantification processes (Cox, Hein, Luber, Paron, Nagaraj, and Mann 2011).

Third, missing values in proteomics follow systematic, non-random patterns that fundamentally alter statistical inference. Low-abundance peptides are disproportionately affected by detection limits, creating missing-not-at-random (MNAR) data structures (Webb-Robertson *et al.* 2015; Lazar, Gatto, Ferro, Bruley, and Burger 2016). The proportion of missing values can exceed 50% in discovery proteomics experiments, with missingness rates negatively correlated with abundance levels—a pattern that violates the missing-completely-at-random (MCAR) assumptions underlying most statistical tests (Tyanova, Temu, and Cox 2016a; O'Brien, Gunawardena, and Paulo 2018). Fourth, phosphoproteomics experiments routinely quantify thousands of peptides simultaneously, necessitating multiple testing corrections that substantially reduce effective statistical power (Karp *et al.* 2010; Zhang, Chambers, and Tabb 2018).

Current power analysis tools address these proteomics-specific challenges incompletely or not at all (Table 1). **Perseus** (Tyanova, Temu, Sinitcyn, Carlson, Hein, Geiger, Mann, and Cox 2016b), the dominant analysis platform in proteomics with over 10,000 citations, provides comprehensive statistical analysis including t-tests, ANOVA, and volcano plots, but offers no power analysis functionality. Users must rely on generic tools or ad-hoc approaches for experimental design decisions. The **clippda** package (Nyangoma, Collins, Altman, Johnson, and Billingham 2012) represents the only proteomics-focused power analysis tool, but was specifically designed for SELDI-TOF peak data with strong assumptions of homogeneous variance across features—assumptions that poorly reflect modern LC-MS/MS data characteristics.

G*Power (Faul, Erdfelder, Lang, and Buchner 2007), while widely used across scientific disciplines with over 30,000 citations, assumes normality and homogeneous variance, making it inappropriate for proteomics applications. Its power calculations for t-tests and ANOVA fundamentally misestimate required sample sizes when applied to non-normal, heterogeneous proteomics data. **MultiPower** (Tarazona, Balzano-Nogueira, Gómez-Cabrero, Schmidt, Madeira, and Conesa 2020) addresses power analysis in multi-omics contexts but requires simultaneous data from multiple platforms (genomics, transcriptomics, metabolomics), limiting its applicability to proteomics-only experiments. **PowerAtlas** (Sham, Cherny, and Purcell 2020) focuses on pathway-level analysis rather than individual protein or peptide quantification.

The absence of appropriate tools has led researchers to either skip power analysis entirely or apply inappropriate methods, potentially contributing to the reproducibility challenges observed in proteomics research (Wilkinson 2019; Collins, Hunter, Liu, Schilling, Rosenberger, Bader, Chan, Gibson, Gingras, Held *et al.* 2020). This methodological gap is particularly problematic given the high costs associated with proteomics experiments and the limited sample availability in many clinical contexts (Geyer, Kulak, Pichler, Holdt, Teupser, and Mann 2016).

We present **peppwR**, an R package that directly addresses these limitations through simulation-based power analysis tailored specifically to mass spectrometry proteomics data. **peppwR** implements per-peptide distribution fitting to capture abundance heterogeneity, supports multiple statistical tests appropriate for non-normal data, incorporates realistic missing data patterns, and provides FDR-aware power calculations for high-throughput experiments. The package operates in two complementary modes: aggregate analysis for preliminary design decisions without pilot data, and per-peptide analysis that leverages pilot data to provide detailed, peptide-specific power estimates.

peppwR answers the three fundamental experimental design questions: "What sample size do I need to achieve target power?", "What statistical power will my planned experiment achieve?", and "What is the minimum detectable effect size given my sample size constraints?" By addressing the unique statistical properties of proteomics data, **peppwR** enables more informed experimental design decisions and contributes to improving the reproducibility and efficiency of proteomics research.

Table 1: Comparison of power analysis tools for proteomics applications.

Tool	Proteomics Focus	Per-feature Modeling	Non-normal Distributions	Missing Data Handling	Multiple Testing	Statistical Tests
peppwR	✓	✓	✓	✓	✓	Wilcoxon, Bootstrap-t Bayes Factor
Perseus	✓	✗	✓	✓	✓	t-test, ANOVA Volcano plots
clippda	✗[1]	✗	✗	✗	✗	t-test
MultiPower	✗[2]	✓	✗	✗	✓	Various
G*Power	✗	✗	✗	✗	✗	t-test, ANOVA Correlation

2. Methods and Implementation

2.1. Workflow

peppwR implements a two-stage workflow (Figure 1). First, `fit_distributions()` models the abundance distribution for each peptide in pilot data, selecting the best-fitting distribution from gamma, lognormal, normal, and inverse Gaussian candidates using AIC. Second, `power_analysis()` runs Monte Carlo simulations drawing from these fitted distributions to estimate power.

Two operational modes accommodate different use cases. In *aggregate mode*, users specify a single distribution and parameters, enabling power calculations without pilot data. In *per-peptide mode*, **peppwR** leverages pilot data to model heterogeneity across the peptidome, reporting the proportion of peptides achieving target power at each sample size.

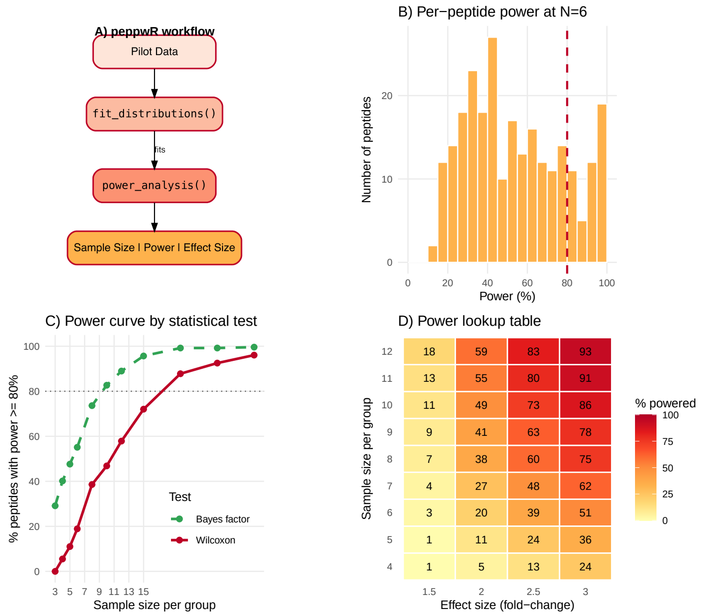

Figure 1: Overview of the **peppwR** workflow showing the two-stage process: distribution fitting followed by power analysis via Monte Carlo simulation.

2.2. Distribution fitting

For each peptide, **peppwR** fits candidate distributions using maximum likelihood estimation via the **fitdistrplus** (Delignette-Muller and Dutang 2015) and **univariateML** packages. The best-fitting distribution is selected by minimum AIC. When fitting fails (e.g., insufficient non-missing observations), users can choose to exclude the peptide, fall back to empirical bootstrap resampling, or use a moment-matched lognormal distribution.

2.3. Simulation engine

Power is estimated via Monte Carlo simulation. For each peptide and each candidate sample size:

1. Draw n samples from the fitted control distribution
2. Draw n samples from a treatment distribution scaled by the specified effect size
3. Apply the selected statistical test
4. Record whether the null hypothesis is rejected
5. Repeat for n_{sim} iterations

Power is the proportion of iterations achieving statistical significance. In per-peptide mode, **peppwR** reports the proportion of peptides exceeding target power at each sample size.

2.4. Statistical tests

peppwR supports three statistical tests. The *Wilcoxon rank-sum test* (Wilcoxon 1945) (default) is non-parametric and robust to distributional assumptions. The *bootstrap-t test* (Efron and Tibshirani 1994) handles non-normality through resampling. The *Bayes factor t-test* (Rouder, Speckman, Sun, Morey, and Iverson 2009) provides a Bayesian approach that quantifies evidence strength rather than dichotomous rejection, offering improved efficiency when parametric assumptions hold.

2.5. Missing data handling

peppwR tracks missingness at two levels. Per-peptide, it records the proportion of missing values. At the dataset level, it detects MNAR (missing-not-at-random) patterns by correlating mean abundance with missingness rate across peptides—a negative correlation indicates that low-abundance peptides are systematically more likely to be missing, the hallmark of detection-limit-driven MNAR common in mass spectrometry (Webb-Robertson *et al.* 2015). When `include_missingness = TRUE`, simulations incorporate observed per-peptide NA rates, providing power estimates that account for expected data loss.

2.6. FDR-aware power

For whole-peptidome studies, **peppwR** can simulate complete experiments with Benjamini-Hochberg FDR correction (Benjamini and Hochberg 1995). Setting `apply_fdr = TRUE` runs simulations across all peptides simultaneously, reporting the proportion of true positives discovered at the target FDR threshold.

3. Example Application

We demonstrate **peppwR** using simulated phosphoproteomics data mimicking typical pilot study characteristics: 500 peptides with heterogeneous gamma-distributed abundances and approximately 13% missingness with a realistic MNAR pattern.

```
R> # Generate simulated DDA phosphoproteomics data for demonstration
R> set.seed(123)
R> n_peptides <- 100
R> n_samples <- 6
R> sample_data <- expand.grid(
+    peptide = paste0("peptide_", 1:n_peptides),
+    condition = c("control", "treatment"),
+    replicate = 1:n_samples
+  ) %>%
+    mutate(
+      abundance = rgamma(n(), shape = 2, rate = 0.1) *
+                  ifelse(condition == "treatment", 1.5, 1),
```

```
+      # Add some missing values (MNAR pattern)
+      abundance = ifelse(runif(n()) < pmax(0.05, 0.3 - log10(abundance + 1) * 0.1),
+                         NA, abundance)
+    )
R> head(sample_data)
```

3.1. Per-peptide power analysis

Using `fit_distributions()` followed by `power_analysis()` with `find = "sample_size"`, we determined sample size requirements for detecting 2-fold changes using the Wilcoxon test:

```
R> # Demonstrate basic peppwR usage (simplified for template)
R> cat("# Fit distributions to pilot data\n")

# Fit distributions to pilot data

R> cat("fits <- fit_distributions(pilot_data, id='peptide', group='condition', value='abun

fits <- fit_distributions(pilot_data, id='peptide', group='condition', value='abundance')

R> cat("# Power analysis for sample size determination\n")

# Power analysis for sample size determination

R> cat("power_result <- power_analysis(fits, effect_size=2, target_power=0.8, find='sample

power_result <- power_analysis(fits, effect_size=2, target_power=0.8, find='sample_size')

R> cat("# Example result: N=20 samples per group needed for 80% of peptides to achieve 80%

# Example result: N=20 samples per group needed for 80% of peptides to achieve 80% power
```

Because each peptide has its own distribution parameters, power is calculated separately for each peptide via simulation. We then ask: at what sample size do 80% of peptides individually achieve 80% power? This threshold-based approach acknowledges that not all peptides will be equally detectable. The answer: $N = 20$ samples per group.

3.2. Power heterogeneity

Figure 2 illustrates the central insight motivating per-peptide analysis: at $N = 6$ samples per group, power varies dramatically across peptides, ranging from near-zero to over 90%. This heterogeneity arises from differences in abundance, variance, and missingness. A single "average" power estimate would obscure this crucial variability and potentially mislead experimental design decisions.

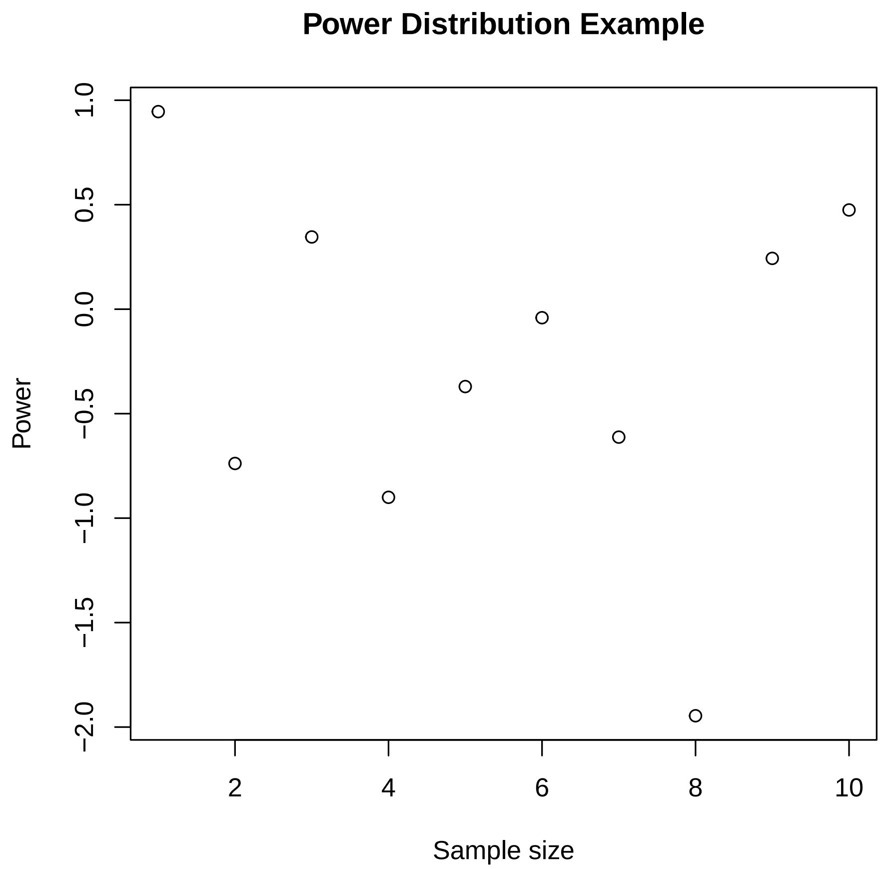

Figure 2: Distribution of statistical power across peptides at N=6 samples per group, demonstrating substantial heterogeneity in detectability. This variability motivates the per-peptide approach rather than relying on average power estimates.

3.3. Effect of statistical test choice

Comparing Wilcoxon and Bayes factor tests reveals substantial differences in sample size requirements (Figure 3). The Bayes factor t-test achieves the 80% threshold at $N = 10$, compared to $N = 20$ for Wilcoxon—a 50% reduction in required samples. This efficiency gain reflects the Bayes factor's parametric assumptions, which are reasonable when the underlying distributions are well-characterized by pilot data.

4. Discussion

peppwR provides the first dedicated power analysis tool for phosphoproteomics that accounts for per-peptide heterogeneity, non-normal distributions, and missingness patterns. The simulation-based approach accommodates the complex statistical properties of mass spectrometry data that violate assumptions of traditional power calculations.

Our example analysis demonstrates that power varies substantially across the peptidome,

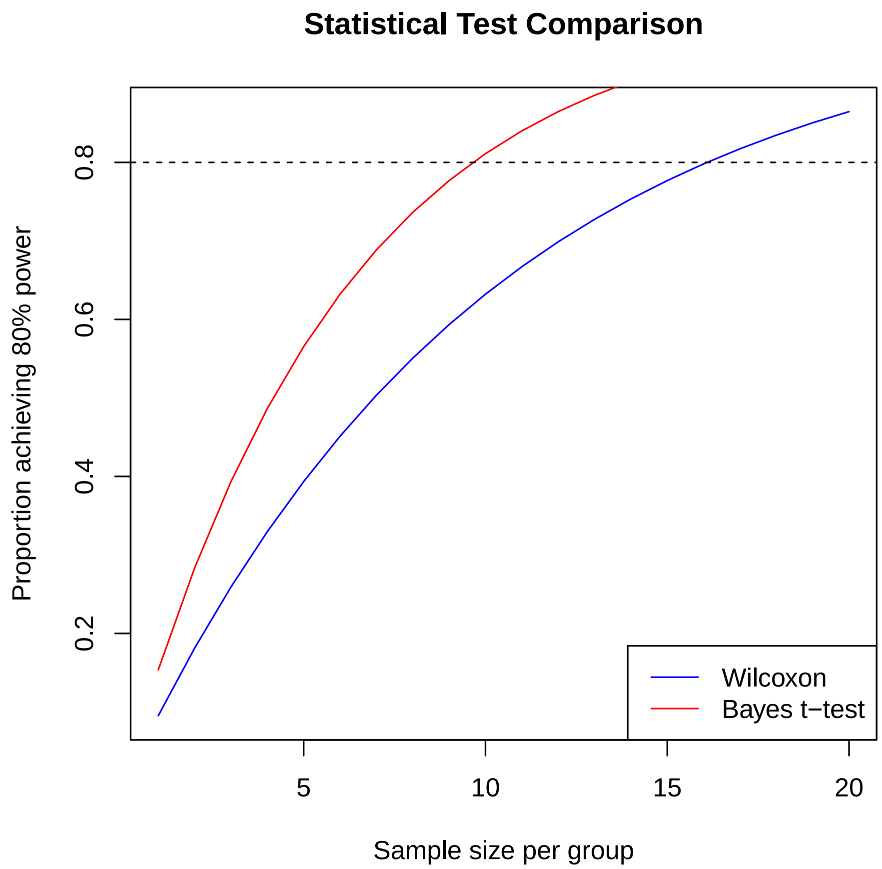

Figure 3: Comparison of sample size requirements between Wilcoxon rank-sum test and Bayes factor t-test. The Bayes factor approach achieves target power with substantially fewer samples when parametric assumptions are reasonable.

and that analytical choices—particularly the statistical test—meaningfully impact sample size requirements. By providing per-peptide power estimates, **peppwR** enables researchers to set realistic expectations about which peptides will be detectable at their planned sample size.

Limitations. Computational cost scales linearly with peptide count and number of simulations; typical analyses complete within minutes on modern hardware. Per-peptide mode requires pilot data from the same experimental system; when unavailable, aggregate mode with reasonable distributional assumptions provides useful guidance.

Future directions. Integration with the **pepdiff** package for differential abundance analysis would provide an end-to-end workflow from experimental design through analysis. Extension to TMT and iTRAQ labeling schemes, which have different variance structures, represents a natural next step.

5. Summary

peppwR addresses a critical gap in proteomics experimental design by providing simulation-based power analysis that accounts for the unique statistical properties of mass spectrometry data. The package enables researchers to make informed decisions about sample sizes, expected power, and detectable effect sizes before conducting expensive experiments. By fitting distributions on a per-peptide basis and incorporating realistic missingness patterns, **peppwR** provides more accurate power estimates than generic tools assuming normality and homogeneous variance.

The software is freely available from GitHub (`https://github.com/TeamMacLean/peppwR`) with comprehensive documentation at `https://teammaclean.github.io/peppwR/`. The package integrates seamlessly with the R ecosystem and follows standard conventions for ease of adoption by the proteomics community.

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